

Redesign Accuracy Roadmap

RD-M1-RD-M5 | Original five-stage plan

SOURCE-GROUNDED PLANNING EDITION

Prepared 06 October 2026. Milestone numbering and scope are preserved. This document is a roadmap, not milestone acceptance or permission to dispatch paid requests.

Current position

- This preserves the original five phases. Later phases are planned work, not completed implementations.
- RD-M1 and RD-M2 evidence exists in the merged project; verify the final independent disposition and installed code before starting the next stage.

Sequence

- RD-M1 - Baseline & Error Inventory
- RD-M2 - Safe Apply & AutoCAD Block Library
- RD-M3 - Deterministic Geometry, Placement & Coordination
- RD-M4 - Multi-Stage AI & Visual Review
- RD-M5 - Real-Project Accuracy Validation & Release

Baseline & Error Inventory

Freeze the current Redesign behaviour and diagnose every observed defect before changing placement logic.

Work scope

- Snapshot code, data and source/output drawings with hashes.
- Map Plan, geometry, AI, review and Apply paths; register Golden Cases and locate each error.
- Separate confirmed causes from hypotheses and identify current coordination and provenance limits.

Deliverables

- Baseline package, pipeline map, failure inventory, Golden Cases and accuracy measures.

Acceptance gate

- A fresh reviewer can reproduce the evidence and trace every finding to its source.
- No algorithm, original drawing or live data is changed by this evidence milestone.

Boundary and dependency

- RD-M1 is documented in the merged source. Do not equate documentation alone with a new acceptance decision.

Safe Apply & AutoCAD Block Library

Generate verified DWG copies from explicitly approved changes and refuse incomplete or unsafe execution.

Work scope

- Resolve CT1/CT2/CR from the current installation library rather than another user's stored path.
- Require approvals and engineer spot confirmation where placement uncertainty exceeds the established threshold.
- Fail closed on script errors; validate standalone completion markers, saved output and read-back reconciliation.
- Use unique output names, cancellation, idempotency and stale-approval checks; never overwrite source or publish to the archive implicitly.

Deliverables

- Candidate patch, unit/regression tests, isolated real AutoCAD evidence and data-safety report.

Acceptance gate

- Successful insert/erase cases reconcile to approved changes; missing blocks and failed steps publish nothing.
- Independent review clears the final correction; integration is separately verified.

Boundary and dependency

- RD-M2 packages and correction evidence exist. The supplied handoff says RD-M3 is not started or authorized by RD-M2.

Deterministic Geometry, Placement & Coordination

Establish trustworthy visible geometry before using AI to choose among valid equipment locations.

Work scope

- Start with the documented bounded wall-index proposal: effective layers, block inheritance, visibility and project wall-layer allow-lists.
- Then stage coordination rules for doors, ceilings, obstacles, containment, rotation, spacing and coverage.
- Use physical equipment/symbol geometry as the anchor, not the position of its text label.
- Re-check geometry and clashes after each proposed change; expose unresolved cases.

Deliverables

- Verified geometry index, rule contracts, collision/coverage evidence and comparison renders.

Acceptance gate

- No placement is justified by invisible or non-wall geometry.
- Rules pass reviewed Golden Cases, including a simple sheet and a rotated view; known failures improve without new critical clashes.

Boundary and dependency

- The first bounded RD-M3 task in the source does not change AI, Apply or all coordination rules at once. Expand only within reviewed stage tasks.

Multi-Stage AI & Visual Review

Use more AI where it improves accuracy, while keeping measurable geometry and engineer authority explicit.

Work scope

- Read the drawing and relevant disciplines; generate only geometrically valid candidate positions.
- Use AI to interpret difficult layouts and choose/explain proposals with source evidence.
- Render proposals, run a separate visual review and perform bounded repairs followed by rule re-checks.
- Preserve uncertainty; require engineer approval before Apply and cap repair rounds to prevent endless loops.

Deliverables

- AI task contracts, structured proposals, render-review evidence, bounded repair history and review UI.

Acceptance gate

- Visual and deterministic checks agree on acceptability; uncertain or conflicting proposals remain held.
- Accuracy improvement is demonstrated against a frozen baseline with full call, latency and failure accounting.

Boundary and dependency

- AI does not establish engineering approval, dimensions or collision clearance by assertion alone. Paid validation needs a declared authorized scope.

Real-Project Accuracy Validation & Release

Validate generalization on real projects and close critical errors before wider use.

Work scope

- Use independently reviewed fresh Golden Cases across layouts, rotations, disciplines and project types.
- Compare before/after placement, coordination, coverage, visual quality, Apply reliability and elapsed time.
- Count missed/held equipment and failed drawings; evaluate repeat-run variation and all new false acceptances.
- Package release, backup/rollback, operational limits and engineer acceptance before rollout.

Deliverables

- Fresh validation results, complete defect disposition, independent review and controlled release plan.

Acceptance gate

- Predeclared accuracy and safety gates pass; no unresolved critical error is promoted.
- No claim of general improvement rests solely on one exposed drawing or one layout.

Boundary and dependency

- This is the final stage of the original five-stage Redesign roadmap. No RD-M6 or RD-M7 is introduced.

Execution and evidence

Accuracy priorities

- Correct physical equipment location, complete equipment coverage, correct orientation, coordination with other disciplines and reliable approved-only DWG output.
- Text labels are evidence for identity, not automatic equipment coordinates. Schedules corroborate equipment; they must not create a second copy of the same physical item.

Execution discipline

- Keep source drawings, live databases and running services separate from candidate validation. Use traceable copies, hashes and explicit promotion.
- Fable: orchestration, Medium effort. Opus: engineering implementation and visual interpretation, High effort. A separate fresh reviewer checks the package.
- Prioritize accuracy over minimizing AI calls, but use bounded budgets and repair loops with visible stops.

Sources and status limits

- Original five-stage plan: the owner's recorded Redesign discussion.
- Detailed evidence: docs/milestones/redesign/RD-M1; RD-M2/NEXT-MILESTONE-HANDOFF.md; RD-M2 reports and independent/correction review packages.
- The detailed RD-M3 wall-index task is sourced from the handoff. RD-M4 and RD-M5 summarize the original planned direction and do not assert implementation or authorization.